

Nathan Ma

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Mechanical Engineering

Sep 2024 - Present

GPA: 3.95 (93.4%)

TECHNICAL SKILLS

Software: SolidWorks, Siemens NX, AutoCAD, Catia 3DX, Ansys, Excel

Documentation: Mechanical Engineering Drawings, Report Writing, GD&T, Gantt Charts

Prototyping & Build Skills: Machining, 3D Printing, Laser Cutting, Soldering

Programming: Python, C++, MATLAB, Arduino, RobotC, LaTeX

EXPERIENCE

General Motors | *Industrial Engineering Co-op*

May 2025 - Aug 2025

- Led a **continuous improvement** initiative focused on eliminating production inefficiencies through targeted **root cause analysis**, collaborating with **cross-functional teams** to optimize line balance. Successfully reduced job cycle times by up to **11.8%** and eliminated significant line downtime, resulting in an increased throughput of Chevrolet Silverado pickup trucks.
- Conducted **time studies** and redesigned standardized work across **30+** stations, utilizing **lean manufacturing** principles to identify wasted movement and ergonomic risks, resulting in noticeable cycle time improvements and reduced ergonomic strain.
- Updated **10,000+ sqft** of factory layout using **AutoCAD**, streamlining cross-functional operations by delivering precise, standardized floor plans that enhanced coordination and efficiency across departments.
- Developed practical understanding of **DFMA** by attending current and future product design review meetings and observing how part design decisions influenced cost, ergonomics, and error-proofing on the plant floor—connecting design intent with real-world assembly outcomes.
- Developed an **Excel-based** readiness document mapping job elements to their associated option codes, tooling, part numbers, and rack sizes, improving efficiency of plant floor changeovers.

UW Midnight Sun Solar Car Team | *Mechanical Engineer*

Sep 2025 – Present

- Currently working on new rotor design under extreme dimensional and DFMA constraints.

PROJECTS

Internal Cycloidal Actuator | *SolidWorks, 3D-Printing, Soldering*

- Engineered a **compact 7:1 cycloidal actuator** optimized for **torque density and efficiency**, applying design principles for dynamic robotic joints.
- Optimized gear and housing tolerances to reduce backlash while preserving backdrivability and achieving smooth, reliable motion.
- Developed a **SolidWorks CAD assembly** of the internal cycloidal gearbox and housing in **SolidWorks**, ensuring **DFMA** and allowing agile design iterations.
- Integrated with a **Moteus BLDC controller**, achieving smooth, precise **closed-loop motion control**.

Pancake Printer | *SolidWorks, 3D-Printing, C++*

- Built a **functional 2D gantry “pancake printer”** by integrating Tetrix structural components, LEGO EV3 motors, and custom 3D-printed adapters.
- Designed a CAD assembly in **SolidWorks**, ensuring mechanical fitments and reducing prototyping time.
- Programmed motion routines in C++, achieving precise, synchronized XY motion for accurate batter extrusion.

Water Rocket Parachute Deployment Device | *SolidWorks, SolidWorks Flow Simulation, 3D-Printing*

- Designed a nose cone attachment in **SolidWorks** to enable parachute deployment at apogee.
- Ran CFD simulations in **SolidWorks Flow Simulation** to refine aerodynamic geometry and reduce drag.
- Created and tested a **low-cost timer-based release mechanism**, calibrated through multiple launches to ensure consistent parachute deployment and safe landings.
- Documented design process in a technical report for mechanical engineering design course.